HAT-P-32b

R-0200

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-32_241130` · created 2026-07-16T22:19:38+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460644.7282 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.054
Transit depth (Rp/Rs)^2	3.44 +/- 2.0 [%]
Orbital Inclination (inc)	89.86 +/- 1.72
Airmass coefficient 1 (a1)	1.004 +/- 0.082
Airmass coefficient 2 (a2)	-0.02 +/- 0.19
Scatter in the residuals of the lightcurve fit is	8.15 %
Best Comparison Star	#1 - [389, 307]
Optimal Aperture	3.52
Optimal Annulus	6.89
Transit Duration (day)	0.1209 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.186 ± 0.054 vs 0.1489 (deviation 0.61σ)
Duration fit vs published	0.1209 d vs 0.1304 d (deviation 1.53σ)
Mid-time O–C	10.9 min
Depth SNR	3.1
Residual scatter	8.15 %

Light curve

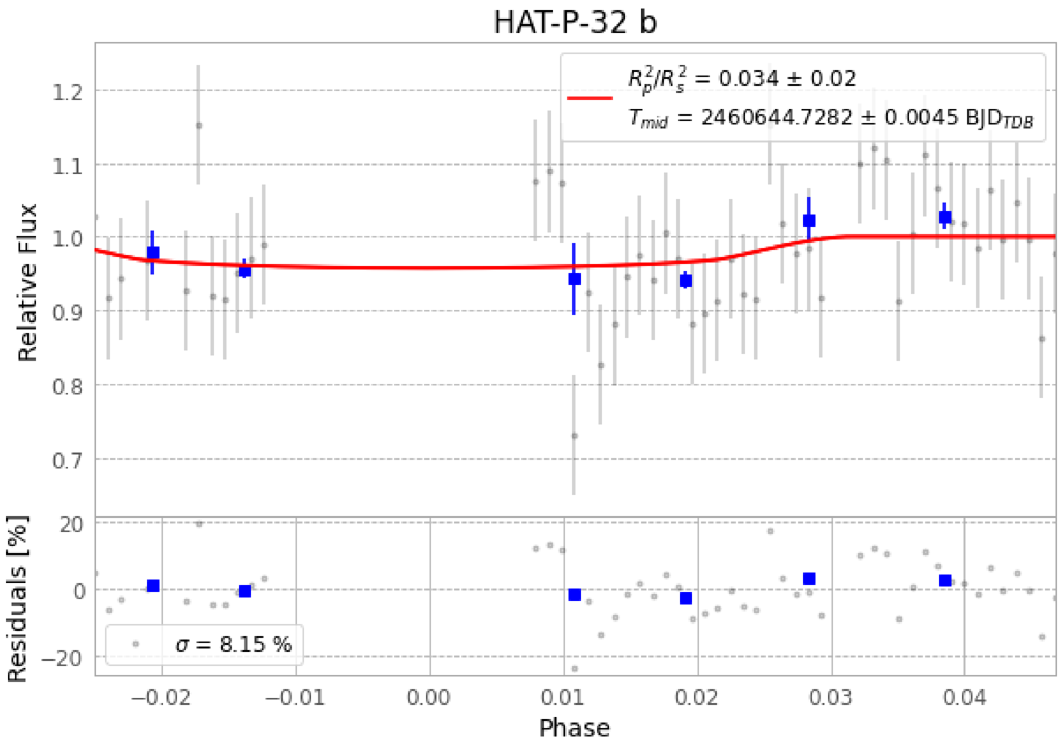

Figure 1 — FinalLightCurve_HAT-P-32 b_2024-11-29.png (SHA-256 9eb9e2ee0b790f3e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 250], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 4a85eb346d6e5e3f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e`

Data provenance

75 FITS frames (49 MB), `HATP-32241130040315.FITS` → `HATP-32241130074515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-241130.log (6740b7c1b97c…), FinalLightCurve_HAT-P-32 b_2024-11-29.png (9eb9e2ee0b79…), FinalLightCurve_HAT-P-32 b_2024-11-29.csv (5eaa36fbe206…)

Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)